

A RESEARCH PROGRAMME IN BOUNDED SELF-MONITORING

SĪDUS

Towards Metarepresentational Ecology

DOCUMENT II

Research Programme

Local Laws, Experiments, and Falsification

司量道

specialist · quantity · way

Truth, maximally sought. Ambition tempered by humility.
Beauty tempered by goodness. Honesty above all else.

Sīdus: Towards Metarepresentational Ecology

II. Research Programme: Local Laws, Experiments, and Falsification

Working research programme - August 2026

Status: defeasible programme; falsifiable local models. This document does not assert that a metarepresentational ecology exists.

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0. Programme rule

The programme is not allowed to infer a global ecology from vocabulary.

It proceeds in the opposite direction:

define a construct → build a discriminating experiment
 → validate a local relation → replicate/generalize
 → only then infer larger organization

The central working hypothesis is deliberately weak:

Some internal monitoring systems may stand in structured, causally important relations to one another. Those relations can be tested one at a time. If sufficiently general regularities emerge, an ecology-level description may become warranted.

Sīdus is the name of the programme while this is the most viable empirical direction. It is not a commitment to preserve any particular legacy construct, experiment, notation, or attention-centric hypothesis.

1. What is inherited from the earlier Sīdus package

Retained

The previous package did several things well enough to keep:

1. **Local identifiability over global latent architecture.** The old unconstrained interaction matrix was correctly retired.
2. **Capacity-only claims.** No result about internal monitoring licenses a phenomenality claim.
3. **The information seam as an experimental scaffold.** Restricted access can create an identifiable communication problem without being asserted as literal neuroanatomy.
4. **Rate spent versus information delivered.** $r_v \neq m_v$ remains a load-bearing distinction.
5. **Joint benefit surfaces.** Complementarity and substitution should be measured rather than assumed away.
6. **Causal use.** Decodability is insufficient.
7. **Optimization validity.** Posterior/channel collapse and training history can masquerade as scientific allocation preferences.
8. **Role/implementation separation.** Functional claims should survive implementation controls where technically feasible.

9. **Stage gates and kill criteria.** Expensive methods and broader claims are conditional on simpler pilots.

Demoted or revised

1. **Attention privilege.** Attention becomes one candidate control process, not the privileged residual target of the programme.
 2. **The old definition of metarepresentation.** A message about an internal controller is now called an *internal-state model* unless stricter representational criteria are independently satisfied.
 3. **One global coordinator.** Π is a tractable experimental consumer, not a theory of cognitive headquarters.
 4. **One global resource currency.** A single rate budget is a controlled experimental manipulation, not a claim about one biological pool.
 5. **Astrocyte as default method.** Gaussian-process Bayesian optimization is optional infrastructure, not a prerequisite for the first experiment.
 6. **Doerig as the programme's ontology.** Doerig-style adversarial pressure is retained; consciousness-specific inferential criteria are not imported where phenomenality is no longer the explanandum.
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2. What would count as an ecology-level result?

A single second-order signal does not make an ecology. Neither do three modules drawn in a diagram.

An ecology-level result requires at least one **relational regularity** that predicts how one internal model changes the value, fidelity, use, or resource allocation of another across more than one condition.

Candidate relation classes are:

- **competition** - models trade off under a demonstrably shared constraint;
- **complementarity** - one model increases the marginal utility of another;
- **substitution** - one model reduces the marginal utility of another;
- **causal-topological dependence** - monitoring value tracks causal role/position rather than implementation identity;
- **recursive regulation** - a model of a control process changes subsequent control;
- **fidelity dependence** - downstream use scales with calibrated accuracy, not merely decodability;
- **temporal niche** - optimal/useful monitoring differs systematically with volatility, persistence, lag, or control horizon.

The programme does not assume all seven exist.

PART 0 - POSITIONING AGAINST ADJACENT LITERATURES

[R4 ·correction, August 2026 review pass]

Constraint C5 of Document I requires that no novelty claim survive an unsearched prior-art row. That constraint was executed thoroughly on the metacognition and consciousness-methodology side and **not** on the information-theory, control, and reinforcement-learning side. Five literatures substantially overlap E1, E3 and E4 and were absent from earlier drafts. Each is given a paragraph here stating what

it established and where this programme actually differs. Where the difference is thin, that is said. A sixth family —the nearest neurocomputational neighbours —was still missing after that pass and is added as P6 in the second external review pass.

P1. Emergent communication in multi-agent RL under bandwidth constraints

Learned communication protocols over limited channels are an established subfield. DIAL and RIAL (Foerster et al. 2016) learn discrete and differentiable protocols end to end; SchedNet (Kim et al. 2019) learns *which* agents may transmit over a contested channel by learned importance weights; IMAC (Wang et al. 2020) derives a mutual-information upper bound as an explicit bandwidth constraint and learns low-entropy messages; VBC (Zhang et al. 2019) prunes communication by variance. E1 —learned agents allocating a shared information rate across sources, with messages tested for causal use —is structurally close to this work, and the resemblance should be stated before a reviewer states it. Three differences are defensible. The sources are **internal control states of a single agent** rather than other agents' observations of external world state, so encoder and consumer share a body and a reward. The **scientific object is the allocation law measured against an empirically estimated benefit surface**, rather than task return maximised by any protocol that works. And **causal-use validation is a gate rather than a diagnostic**: a decodable but behaviourally inert channel fails E0-A outright. Honesty about the first difference: a single agent with private modules and a shared reward is formally close to a cooperative multi-agent system, and the architectural distinction alone will not carry the positioning. The scientific one will.

P2. Information-theoretic bounded rationality

Ortega and Braun's free-energy formulation, and Genewein et al. (2015) on hierarchical decision-making under an information-processing cost, establish the trade-off between expected utility and rate as an optimality principle, with antecedents in Tishby and Polani (2011) and in linearly-solvable control. The r versus m distinction and the KL rate cost used throughout this package live **inside** this tradition and are inherited from it, not derived here; they should be cited accordingly. The difference is one of direction. That literature specifies the trade-off normatively and derives optimal policies from it. E1 applies the constraint to self-monitoring specifically and **measures** the benefit surface empirically rather than assuming a parametric form for it —which is what permits interaction, non-concavity, and failure of separability to appear as findings rather than as violated assumptions.

P3. Rational inattention

Sims (2003) formalised capacity-constrained attention allocation under a Shannon cost; Caplin and Dean (2015) supplied revealed-preference tests for costly information acquisition. H1 is, in structure, a rational-inattention hypothesis pointed inward, and this is the collision most likely to be recognised by a cognitive scientist rather than only by a machine-learning reviewer. Two differences. Rational inattention constrains the acquisition of **external** state; here the constrained resource is applied to states the agent itself generates, where the benefit accrues to control of control rather than to decision accuracy. More importantly: rational inattention **assumes the optimum is attained** and derives behaviour from that premise. This programme treats attainment as the empirical question. The real content of H1 and H2 is not whether a capacity-constrained optimum exists —it does —but whether a learning system operating under realistic optimisation pathologies finds it, and what it does instead when it does not.

P4. Value-of-information sensor scheduling and networked control

“Which sensor’s state is worth transmitting under a communication constraint” has been formalised for decades. Imer and Başar (2005), Lipsa and Martins (2011), and Gao, Akyol and Başar (2018) treat remote estimation in which communication is limited or explicitly priced; Gao et al. directly distinguish soft transmission costs from a hard limit on transmission opportunities. Sequential rate-distortion work by Tanaka et al. (2017), and later finite-rate sequential source coding by Stavrou, Skoglund and Tanaka (2022), formalises closely related rate-distortion and encoder-design problems for dynamical systems. **Some E1 specialisations are formally very close to this problem class**, and pretending otherwise would be indefensible; the package does not claim a general isomorphism without an explicit mapping of state, information pattern, objective and constraint. The differences are that the intended sources are non-Gaussian and low-cardinality internal control states, the distortion analogue is expected return rather than quadratic estimation error, and the encoders are **learned rather than designed**. The right posture toward this literature is not defensive. Per Document I §8.1, a deliberately constructed solved linear-Gaussian case is not a competitor but a **validation target**: a learned allocation that cannot recover a known optimum where one exists has not earned interpretation where none does.

P5. Rational metareasoning and value of computation

Russell and Wefald (1991) and Hay, Russell, Tolpin and Shimony (2012) price a computation by its expected effect on the decision it informs. The hypothesis that monitoring value tracks causal position and downstream reach —**H2** in this document, **H4** in the Master’s scoping document — overlaps conceptually with value of computation, and the overlap must be stated. Graph position and value of computation are not identical, but they are close enough that a topology result could be a VOC result wearing different notation.

[Revised after release-candidate review.] An earlier version of this paragraph described VOC as “myopic and per-item” and contrasted that with a joint, constrained Sidus problem. That characterisation was wrong. Rational metareasoning is formulated as a **metalevel MDP**; the myopic or meta-greedy policy is one *approximation* within that framework, not its definition, and Hay et al. explicitly analyse the relation between myopic and optimal metalevel policies. Exact metalevel policies are intractable in general, which is why myopic approximations are used in practice — a fact that applies symmetrically to $\mathbf{r}^*(C)$ and is already covered by anti-slop rule 5.

The honest differences are narrower. VOC prices **computations against a decision**; the allocation problem prices **channels against each other** under a binding shared budget, where sources may be complementary or substitutable. Neither is tractable in general; both enter as benchmarks rather than as claims about what any system computes. For the separable, non-interacting case they converge, which is why a **specified myopic-VOC baseline** belongs in the comparison set rather than in a related-work paragraph. If the joint resource surface fails to improve out-of-sample prediction over myopic VOC, the experiment provides **no incremental predictive evidence** for the interaction structure this programme is built to detect beyond that simpler benchmark.

P6. Precision weighting, attention-as-resource, and working-memory limits

[Second external review pass —neuroscience-side C5.] The nearest neurocomputational neighbours are not source-coding problems at all, and they constrain a different part of the programme. Expected-precision accounts of attention (Feldman & Friston 2010) treat attention as the optimisation of precision estimates that weight first-order sensory evidence — formally, E2’s consumer using M_v to weight first-order representations is precision weighting, and any neuroscience-facing reading of E2 must say so. Attention-as-resource (Kahneman 1973) is the classical ancestor of every shared-budget intuition in this package. The working-memory slots-versus-resources debate (Ma, Husain & Bays

2014) shows that even for the best-studied candidate “shared pool” in cognition, the existence of a single common currency is contested rather than established. None of this changes the synthetic thesis, which makes no biological budget claim. It becomes load-bearing at E6, where these literatures —not the source-coding apparatus —set the standard of evidence for any human translation.

What the six leave standing

Nothing in the formalism. The rate machinery, the shared-budget structure, the utility-versus-rate trade-off, and the marginal-value logic are all inherited. What survives is narrower and should be stated in exactly these terms:

*The application of an established rate-allocation apparatus to **internal monitoring targets specifically**, with **construct validity gated before** any allocation claim (E0), and with **causal use required rather than decodability**.*

That is an application-level and methodological contribution. It is smaller than “a new theory of bounded self-monitoring” and it is the size that the evidence can currently carry.

PART I - HYPOTHESIS FAMILIES

3. H1 - Scarcity and competition

Hypothesis

When multiple internal targets are locally available but their models must pass through a genuinely binding shared communication constraint, learned systems will allocate unequal rate according to the marginal control value of exposing each target.

The safest empirical object is the joint benefit surface:

$$\mathcal{B}(\mathbf{r}) = \mathbb{E}[J \mid \mathbf{r}] - \mathbb{E}[J \mid \mathbf{0}],$$

with normative benchmark

$$\mathbf{r}^*(C) = \arg \max_{\mathbf{r} \geq 0, \sum_v r_v \leq C} \mathcal{B}(\mathbf{r}).$$

This is not a claim that organisms compute the argmax. It is a benchmark against which learned allocation can be compared.

Competing explanations

- source entropy alone;
- ordinary inferability from public information;
- fixed head/implementation bias;
- training-order/path dependence;
- no shared resource at all;
- a simpler architecture in which every source is transmitted independently.

Falsifying / narrowing outcomes

H1 is weakened or rejected in a given architecture if:

- reducing nominal capacity does not create measurable cross-source trade-offs;
 - allocation differences vanish after ordinary inferability is controlled;
 - rates are dominated by head identity or initialization;
 - learned allocations do not generalize to held-out task conditions better than simpler predictors;
 - transmitted state information is behaviorally unused.
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4. H2 - Causal topology predicts monitoring value

Hypothesis

The value of monitoring an internal target depends partly on its causal position in the system. Holding obvious statistical properties as controlled as practical, states with greater downstream reach, bottleneck position, persistence, or influence over other computations should exhibit systematically different benefit-rate curves.

This reframes the old Sīdus topology model from a nuisance control into a candidate explanatory principle.

Minimal test

Create matched modules from the same implementation family but vary one high-level causal property, such as how many downstream decisions depend on the state. Measure:

$$B_v(r) = J(r_v = r) - J(r_v = 0)$$

and ask whether topology predicts the curve out of sample.

Stronger test

If simple manipulations establish an effect, use causal-role permutation. Only then consider causal-abstraction/interchange validation to verify that a proposed high-level role is actually realized by the low-level module.

Failure condition

If topology metrics add no predictive value beyond entropy, inferability, and direct control relevance, do not elevate topology into an organizing principle.

5. H3 - Recursive monitoring can improve control of control

Hypothesis

Monitoring the *realized state* of a control or resource-allocation process can have future control value when commands are imperfect, the realized state persists or lags, and future actions can compensate.

Use two distinct resource levels:

$$\sum_i a_{i,t} \leq C_{\text{base}}$$

for first-order world-facing allocation, and

$$\sum_v r_v + r_A \leq C_{\text{meta}}$$

for internal-state modelling.

The recursion is temporal:

allocate $\rightarrow$ observe realized allocation $\rightarrow$ use the model $\rightarrow$ allocate better later.

Primary estimand

Let

$$B_A^H(r) = J_H(r_A = r) - J_H(r_A = 0)$$

be finite-horizon benefit. Separate current interpretive value from future feedback value with a matched contrast:

$$\Gamma_A^H(r) = B_A^{\text{feedback}, H}(r) - B_A^{\text{frozen}, H}(r).$$

The frozen condition makes the same current allocator information available but prevents it from influencing future allocation commands.

Predictions

Γ_A^H should increase when:

- command-to-realized allocation is noisy or lagged;
- the realized allocation changes downstream signal reliability;
- allocation has temporal persistence;
- the horizon permits corrective reallocation.

It should collapse when:

- realized allocation is directly observable without the special channel;
- commands determine allocation perfectly and immediately;
- allocation has no behavioral consequences;
- feedback is impossible;
- the horizon is too short.

Prior-art discipline

This experiment lives next to meta-control, rational metareasoning, active information gathering, and oMCD. Its novelty is **open**, not presumed.

6. H4 - Faithful self-models are more than decodable echoes

Hypothesis

A candidate internal model earns a functional claim only when its information is sufficiently calibrated for its consumer and selectively perturbing it changes the downstream computation predicted by its content.

A useful evidence sequence is:

decodable < conditionally specific < misrepresentable
< causally target-dependent < causally used.

Failure condition

If a candidate signal decodes the target but perturbing it leaves the putative consumer unchanged, call it redundant or unused under the tested conditions. Do not rescue the claim by redefining its function.

7. H5 - Complementarity and substitution

Hypothesis

The value of one internal model depends on what other models are available.

For a smooth estimated benefit surface, local cross-curvature can be used as a *diagnostic*:

$$\Gamma_{uv}(\mathbf{r}) = \frac{\partial^2 \mathcal{B}(\mathbf{r})}{\partial r_u \partial r_v}.$$

Positive values suggest local complementarity; negative values suggest substitution. The derivative is not an ontological quantity and should not be interpreted without uncertainty.

More robust empirical alternative

Use factorial allocation contrasts or held-out predictive comparisons between:

- additive model $\sum_v B_v(r_v)$;
- pairwise-interaction model;
- full joint surface.

The simplest model that predicts held-out returns adequately wins.

8. H6 - Temporal niche

Hypothesis

Monitoring policies depend systematically on the timescale of the target and the horizon over which its state matters.

Candidate parameters include:

- target volatility;
- temporal autocorrelation;
- command/actuator lag;
- persistence of errors;
- duration of downstream consequences;
- cost of delayed correction.

Prediction

Fast, volatile targets should favor frequent low-latency updating when they remain control-relevant; slowly drifting but consequential targets may favor lower-frequency, higher-fidelity summaries. The exact pattern is an empirical question.

Failure condition

If timescale manipulations do not improve predictions of monitoring policy beyond static control value, temporal niche remains unnecessary.

PART II - EXPERIMENTAL PROGRAMME

9. E0 - Construct benchmark: what actually counts as meta?

This is the first new experiment required by the reframing. It prevents the programme from building an ecology out of an undefined construct.

Goal

Create synthetic systems with known computational relationships and test whether proposed measurements distinguish:

1. **ordinary first-order transformation;**
2. **internal-state model;**
3. **content/property metarepresentation;**
4. **process metarepresentation.**

Minimal architecture

Let an external latent state X_t generate observation O_t . A first-order system produces

$$R_t = F_{\eta_t}(O_t),$$

where η_t controls a hidden property of the representational process: noise, bias, gain, source type, or mapping rule.

Create four candidate downstream systems:

A. First-order echo

Receives R_t and transforms it for the task. It has no separate target representing a property of R_t .

B. Internal-state monitor

Receives a hidden controller/process state C_t and reports its value.

C. Property metarepresentation candidate

Must infer $\phi(R_t)$, such as the likely correctness, source, or reliability of the current first-order representation.

D. Process metarepresentation candidate

Must infer η_t or a compact property of F_{η_t} , and use that model to predict how first-order representations will change under controlled counterfactual inputs.

Critical matched controls

- hold external X_t constant while changing η_t ;
- hold first-order choice/output constant while changing reliability;
- create cases where confidence/source/process inference is wrong;
- perturb the candidate meta-state while preserving first-order R_t ;
- compare against a shortcut model that predicts η_t from public context without reading the first-order process.

Main question

Which evidence tiers from Foundations actually discriminate the four architectures?

Success

E0 succeeds if at least one operational test distinguishes a genuine property/process-monitoring architecture from first-order and internal-state controls under held-out interventions.

Failure

If proposed criteria cannot distinguish the architectures without appealing to their labels or implementation history, the programme does **not** yet possess an empirically useful construct of metarepresentation. Stop and refine the construct before scaling.

10. E1 - Scarcity terrarium: bounded internal-state monitoring

This is the direct descendant of the strongest old Sīdus experiment, but it no longer pretends all messages are strict metarepresentations.

Question

When several private internal control/process states are useful to a coordinating agent but can only be exposed through a binding shared information-rate channel, what predicts the learned allocation of monitoring capacity?

Environment

Use a compact POMDP with at least three low-cardinality private internal states, for example:

- sensory-selection state;
- working-memory update state;
- action-routing/effector state.

The consumer receives ordinary public information $\mathcal{J}_t$ but no direct read of these private states.

Each source has a symmetric encoder:

$$C_{v,t} \rightarrow S_{v,t} \rightarrow U$$

with a single rate constraint:

$$\sum_v r_v \leq C_{\text{meta}}.$$

Required measures

For each source:

- conditional uncertainty $H(C_v \mid \mathcal{I})$;
- state entropy/cardinality;
- causal return sensitivity;
- isolated learnability;
- distortion-rate / benefit-rate curve;
- temporal persistence/update frequency;
- basic causal-topological measures;
- rate spent r_v ;
- incremental information delivered m_v ;
- targeted-channel causal-use effect.

Required controls

- direct-read oracle;
- no-seam/full-read control;
- non-binding budget control;
- source/head permutation across seeds;
- warm-up before scarcity;
- gradual constrained scarcity ramp;
- selected from-scratch scarcity runs;
- held-out task conditions.

Constraint implementation: Arm S core, alphabet arm optional

[Design addition, August 2026 —demoted after release-candidate review]

An earlier version of this section proposed a hard alphabet-cardinality budget as a mandatory second arm and made **rank-order invariance across arms** a confirmatory prediction. Both claims were too strong and are withdrawn. What follows is the corrected position.

Arm S is the core experiment. Primal-dual or augmented-Lagrangian training formulates $\sum_v r_v \leq C_{\text{meta}}$ as a constrained optimisation problem rather than merely attaching a fixed penalty coefficient. In a non-convex learned system, however, **feasibility is an empirical property of the trained solution**, not something the optimiser name guarantees. Realised constraint violation must therefore be reported and preregistered feasibility tolerances met before the run is interpreted as genuinely budget-constrained. The second epistemic question is whether the variational surrogate corresponds tightly enough to the intended information currency—and correction R2 makes that looseness measurable:

$$r_v = m_v + \Delta_v^{\text{prior}}, \quad \Delta_v^{\text{prior}} = \mathbb{E}_{\mathcal{I}} D_{KL}(q(S_v \mid \mathcal{I}) \parallel p(S_v \mid \mathcal{I})).$$

Arm S therefore requires two things to be reported alongside every allocation: realised constraint violation, and Δ_v^{prior} per source as a tightness diagnostic. Those reports are what license the surrogate as a rate meter; a second arm is not needed for that purpose.

Why the invariance test does not work. A KL budget and an alphabet budget are different resource currencies:

$$\mathbb{E}[D_{KL}] \neq \log_2 |\mathcal{Z}|.$$

One constrains expected informational departure from a prior; the other constrains maximum representational capacity. A source may be granted a large alphabet and use it sparsely; another may use

a small alphabet efficiently. Different allocation rankings can therefore arise **under perfect optimisation and an unchanged task**. So

$$\text{ranking}_S \neq \text{ranking}_H$$

does not license the conclusion that the allocation law is optimiser-dependent. The more likely reading is that allocation depends on **which resource is scarce**, which is interesting and is not what an invariance test claims to measure.

The alphabet arm as optional robustness. If a stable allocation phenomenon is established in Arm S, a hard-capacity arm becomes worth running as a **currency-sensitivity** experiment: does the ordering change when the scarce resource is capacity rather than expected information cost? Run that way, a divergence is a finding about currency dependence rather than a failed invariance check.

If it is run, the construction matters. Per-source codebooks give each source its own hard cap and abolish competition; a single shared codebook abolishes source-factorisation and with it the per-source identifiability of r_v , m_v and targeted intervention, taking Gates A3 and A4 and H3 with it. The construction that is hard, shared and factorised at once constrains **total alphabet cardinality with a learned split**:

$$\sum_v \log_2 |\mathcal{Z}_v| \leq C_{\text{meta}},$$

with per-source cardinality drawn from a shared slot pool by a differentiable arbitration. That is the right way to build the arm. It is not a reason to build it before the phenomenon exists —anti-slop rule 8 applies to this proposal as much as to any other.

Confirmatory comparisons

At minimum compare:

1. uniform allocation;
2. entropy-only / inferability-only allocation;
3. isolated causal-value predictor;
4. **myopic value-of-computation prediction** (Russell & Wefald 1991; Hay et al. 2012) —the closest competitor drawn from an established formalism; specified as one-step value, not VOC in general; see Document III, H2;
5. joint empirical resource model;
6. joint resource + topology features;
7. implementation/head identity.

Do **not** make attention residual Δ_A the centre of the thesis. Estimate generic source residuals Δ_v if useful.

Publishable outcomes

- stable scarcity law;
- topology improves prediction;
- strong source interactions invalidate additive resource models;
- learning repeatedly fails to discover a known normative value;
- allocation is path-dependent rather than law-like;
- no genuine trade-off exists.

All are informative.

11. E2 - Strict metarepresentation under scarcity

E2 proceeds only if E0 yields a defensible operational distinction and E1 demonstrates a real shared monitoring constraint.

Question

When first-order representations differ in reliability, source, or generative process, does a bounded system allocate scarce rate to models of those representational properties/processes, and does the allocation improve control beyond first-order shortcuts?

Example

Two perceptual experts produce first-order estimates:

$$R_{1,t} = F_{1,\eta_{1,t}}(O_t), \quad R_{2,t} = F_{2,\eta_{2,t}}(O_t).$$

Their hidden process regimes alter reliability. Candidate meta-generators produce:

$$M_{v,t} = G_v(R_{v,\leq t}, \text{process evidence})$$

about $\eta_{v,t}$, source, or correctness probability.

The consumer uses M_v to weight first-order evidence.

Key comparison

Compare against:

- direct reliability cue;
- public-context predictor;
- first-order network with matched parameter count;
- internal-state monitor that knows a controller setting but not the representational process;
- meta-process model.

E2 is where the phrase **metarepresentation under scarcity** becomes substantively earned rather than merely inherited.

12. E3 - Causal topology and role dependence

Proceed only if E1/E2 produce a robust phenomenon.

Phase 1: cheap topology manipulation

Change downstream reach, bottleneck status, persistence, or branching while holding module implementation fixed where possible.

Phase 2: simple role permutation

Swap functional interfaces across matched implementations. Measure whether monitoring benefit follows role or implementation.

Phase 3: causal-abstraction validation, conditional

Only if the role claim is central and simpler validation is inadequate, construct a high-level causal model and test interchange-intervention faithfulness.

A failed abstraction alignment blocks a high-level role conclusion. It does not invite a more elaborate rescue model.

13. E4 - Recursive allocator monitoring

This retains the two-budget S1-R construction because it converts a tempting regress into a temporal causal question.

Crucial terminology:

- if A_t is merely the realized resource-allocation/control state, R_A is an **internal-state model**;
- call it a strict metarepresentation only if the target is explicitly a representational property/process of the allocator.

The primary test is feedback versus frozen feedback, not an unexplained allocator residual.

14. E5 - Two validated meta-systems in interaction

This is the first experiment that would genuinely deserve the phrase **metarepresentational ecology**.

Use two previously validated meta-systems, for example:

- a model of perceptual reliability/source;
- a model of memory reliability or control allocation.

Ask whether their joint utility is additive, complementary, substitutable, or mediated through a third control process.

Minimal criterion for an ecology-level claim

At least one interaction must:

1. survive selective interventions on each component;
2. improve held-out prediction over independent-module models;
3. recur under a second task or architecture;
4. not reduce to shared external input or generic confidence bias.

Until then, use “candidate interaction” rather than “ecological law.”

15. E6 - External translations

Human metacognition

Plausible targets include perception-memory confidence, source monitoring, error monitoring, and meta-control. The literature already warns against assuming strong domain generality. Human work should therefore test specific interactions and trade-offs, not merely group differences or correlations.

Artificial-system introspection

The same construct discipline can test whether a model's report about an internal state is causally faithful rather than semantically plausible. Define the internal target independently; intervene on target and report separately; vary semantic framing; preserve non-target information.

Clinical translation

This remains post-construct-validity. Hypervigilance, interoception, alexithymia or dissociation should not be redescribed as “metarepresentational ecology” unless the framework predicts a distinctive within-person interaction or trade-off that competing accounts do not.

PART III - WHAT CAN AND CANNOT BE COMPUTED

16. Computational tractability map

Object/question	Can render now?	Notes
private low-cardinality internal process state	Yes	direct simulator ground truth
internal-state model of that process	Yes	supervised or learned encoder
rate spent r_v	Yes	variational KL control surrogate
incremental target information m_v	Yes, with validation	low-cardinality targets + held-out calibrated log loss
causal use of message	Yes	selective message intervention, support checks
joint benefit surface	Yes for small K	grid/design first; surrogate only if needed
complementarity/substitution	Yes, locally	factorial contrasts or uncertain surface derivatives
hidden process/reliability regime	Yes	known by simulator
process metarepresentation	Yes in proof-of-concept form	must beat first-order/public-context controls
topology manipulations	Yes in synthetic systems	exact matching may be impossible
causal abstraction	Conditional	high overhead; only after robust role effect
recursive allocator feedback	Yes	requires two budgets and temporal dynamics
one universal cognitive resource currency	No	experimental single-budget model is not neural fact
spontaneous discovery of arbitrary internal ontology	Not currently identifiable without strong assumptions	separate long-horizon problem
complete brain metarepresentational ecology	No	no current measurement/identification route

Object/question	Can render now?	Notes
phenomenal consciousness	No	explicitly outside the programme

17. Astrocyte: optional methods module

Astrocyte remains valid **only when the empirical surface is expensive enough to justify it.**

The scientific target is still:

$$\mathbf{r}^*(C) = \arg \max_{\mathbf{r} \in \mathcal{S}_C} \mathcal{B}(\mathbf{r}).$$

Gaussian-process Bayesian optimization estimates that target; it does not define it.

Escalation rule

Use, in order:

1. small factorial / simplex design;
2. interpolation / regression with uncertainty;
3. GP surrogate if surface smoothness is empirically defensible;
4. noise-aware BO only when each new training configuration is genuinely expensive and active search materially reduces cost.

Do not deploy GP curvature diagnostics or Bayesian optimization because they make the project look sophisticated.

PART IV - FALSIFICATION AND STAGE DISCIPLINE

18. Questions that can actually be false

Question	Evidence for	Evidence against
Does a shared monitoring budget create competition?	cross-source trade-offs appear only when budget binds	no displacement/trade-off under validated scarcity
Does dynamic value predict allocation?	held-out rates/returns better predicted than by entropy/identity baselines	no out-of-sample advantage
Does topology matter?	controlled causal-position changes shift benefit/allocation	topology adds no predictive value
Are messages causally used?	selective perturbation changes predicted behavior with first-order target intact	decodable message can be removed without relevant effect
Is a candidate state meta rather than first-order?	survives E0 conditional, decoupling and intervention controls	simple first-order/public-context model explains data
Does recursive monitoring help?	feedback-vs-frozen benefit appears where lag/noise/horizon predict it	no specific feedback benefit or benefit persists when feedback impossible

Question	Evidence for	Evidence against
Are two meta-systems complementary/substitutable?	robust interaction improves held-out prediction	additive independent model suffices
Is there an ecology-level regularity?	same relational principle generalizes to second task/architecture	every effect is domain/task specific

19. Global kill/narrow criteria

The programme should be narrowed or abandoned as an ecology-level project if repeated work shows that:

- strict metarepresentational constructs cannot be operationally distinguished from first-order/process-reflexive alternatives;
- apparent interactions vanish under conditional-information and common-cause controls;
- shared-resource hypotheses repeatedly fail to generate trade-offs;
- topological features do not generalize;
- candidate models are decodable but not causally used;
- results are dominated by optimization history or implementation identity;
- each domain requires an unrelated explanation and a common relational vocabulary adds no predictive compression.

In that case the correct conclusion is not “the ecology is more complicated.”

It is:

There may be no useful ecology-level theory. Study the domain-specific mechanisms instead.

20. Stage gates

Gate 0 - Construct validity

Proceed only if E0 distinguishes at least one candidate metarepresentational relation from simpler controls.

Gate 1 - Scarcity exists

Proceed beyond E1 only if the shared budget produces measurable, stable trade-offs and messages are useful.

Gate 2 - Relation is not training artifact

Proceed to topology/recursion only if rates and causal effects survive reasonable seed/schedule/implementation controls.

Gate 3 - Local law exists

Use the word “principle” only after held-out prediction and intervention succeed.

Gate 4 - Ecology claim

Use the word “ecology” as an empirical result only after at least two validated monitoring/meta-systems show a reproducible interaction across more than one task or architecture.

21. Immediate sequence

The programme should generate code before another conceptual companion.

1. **Systematic construct/prior-art review:** Proust, Shea, Cleeremans/SOMA, Kanai et al., metacognition architectures, meta-control, resource-rationality, distributed communication.
2. **E0 notebook:** synthetic first-order vs internal-state vs property-meta vs process-meta benchmark.
3. **E1 minimal environment:** no GP, no causal abstraction, no attention residual as primary target.
4. **Pilot stage gate:** prove an information seam, a binding shared budget, stable informative messages, and causal use.
5. **Freeze the smallest successful model.** Preregister/replicate.
6. Only then select **one** extension: topology, strict-meta scarcity, or recursive control.

Not all three.

22. What success would look like

A successful first paper need not mention consciousness or claim a new cognitive architecture.

A strong result could be:

In a controlled modular agent, private internal process states competed for a binding information-rate channel. Learned allocation was predicted by the joint dynamic control benefit of monitoring those states, and transmitted information was causally used. A controlled change in causal position altered monitoring value in the predicted direction.

A stronger later result could be:

A separately validated metarepresentation of a first-order representational process competed with other internal models under the same constraint, and its value depended on the process's uncertainty and causal role.

An ecology-level result would require more:

The same relational principle predicts interactions among multiple validated meta-systems across tasks or architectures.

The programme must earn each sentence separately.

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